# YU-SHIN HUANG

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## EDUCATION

**Texas A&M University**, College Station, TX — Aug. 2024 – Aug. 2028 (Expected)
Ph.D., Electrical Engineering, Information Science and Learning Systems — **GPA: 4.0/4.0**
*Research Areas:* Constrained Optimization, LLM Watermarking and Steganography, Stochastic Processes, Markov Decision Processes, Information Theory
*Relevant Coursework:* Advanced Convex Optimization, Advanced Stochastic Processes, Estimation and Detection Theory, Online Decision Learning, Applied Bayesian Methods, Deep Learning, Real Analysis

**Texas A&M University**, College Station, TX — Aug. 2022 – May 2024
M.S., Electrical Engineering, Signal Processing and Imaging — **GPA: 4.0/4.0**
*Relevant Coursework:* Convex Optimization, Scientific Machine Learning, Pattern Recognition, Wireless Communication, Digital Image Processing

**National Tsing Hua University**, Hsinchu, Taiwan — Sep. 2017 – Jun. 2022
B.S., Mathematics & Electrical Engineering, Double Major — **GPA: 3.4/4.3**
*Relevant Coursework:* Convex Optimization, Queueing Theory, Machine Learning, Communication Systems, Digital Signal Processing, Advanced Calculus, Complex Analysis

## TECHNICAL SKILLS

- **Optimization & Applied Mathematics:** Convex and constrained optimization, Constrained Markov decision processes, stochastic processes, information theory
- **Computational Methods:** Numerical optimization, Monte Carlo simulation, Statistical detection
- **Programming & Tools:** Python, MATLAB, CVX, C, C++, LaTeX

## RESEARCH EXPERIENCE

**Graduate Researcher / Graduate Research Assistant, Texas A&M University** — Aug. 2023 – Present

- **LLM Steganography via Constrained Optimization and CMDPs**
  Developed optimization frameworks for LLM probability-distribution design under statistical constraints, including total-variation and divergence constraints. Extended the framework to sequential generation using constrained Markov decision processes, convex reformulations, and structured optimal policies.
- **Multi-bit Generative Watermarking Under False-Alarm Constraints**
  Formulated and solved a multi-bit generative watermarking optimization problem under worst-case false-alarm constraints. Derived optimal embedding and detection strategies and evaluated performance through theoretical analysis and numerical experiments.
- **Weakly and Leaky Private Information Retrieval**
  Designed optimized probability-allocation and retrieval strategies for private information retrieval under heterogeneous trust and privacy constraints. Developed schemes that reduce information leakage and improve the leakage scaling compared with existing approaches.
- Develop mathematical formulations, derive solution structures and optimization algorithms, and validate theoretical results through numerical experiments and Monte Carlo simulations in Python, MATLAB, and CVX package. Research has resulted in publications at EACL, IEEE ISIT, IEEE ITW, IEEE Transactions on Information Theory, and IEEE Transactions on Information Forensics and Security.

**Undergraduate Researcher, National Tsing Hua University** — Jan. 2021 – Jan. 2022

- Collaborated with National Cheng Kung University on hyperspectral compressed sensing for terahertz imaging and material characterization.
- Developed hardware and software components for the sensing system and contributed to experimental data collection and analysis.

## PUBLICATIONS

- **YS Huang**, C. Tian, and K. Narayanan, "An Optimal Multi-bit Generative Watermarking Scheme Under Worst-Case False-Alarm Constraints," *IEEE Information Theory Workshop (ITW)*, 2026. Extended version submitted to *IEEE Journal on Selected Areas in Information Theory (JSAIT)*.
- **YS Huang**, P. Just, H. Yin, K. Narayanan, R. Huang, and C. Tian, "OD-stega: LLM-based Relatively Secure Steganography via Optimized Distributions," *European Chapter of the Association for Computational Linguistics (EACL)*, 2026.

- **YS Huang**, C. Tian, K. Narayanan, and L. Zheng, "Relatively-Secure LLM-Based Steganography via Constrained Markov Decision Processes," *IEEE International Symposium on Information Theory (ISIT)*, 2025.
- **YS Huang**, W. Zhao, R. Zhou, and C. Tian, "Weakly Private Information Retrieval from Heterogeneously Trusted Servers," *IEEE International Symposium on Information Theory (ISIT)*, 2024.
- W. Zhao, **YS Huang**, C. Tian, and A. Sprintson, "Optimizing Leaky Private Information Retrieval Codes to Achieve $O(\log K)$ Leakage Ratio Exponent," *IEEE Transactions on Information Forensics and Security*, 2026.
- W. Zhao, **YS Huang**, R. Zhou, and C. Tian, "Weakly Private Information Retrieval from Heterogeneously Trusted Servers," *IEEE Transactions on Information Theory*, 2025.

## TEACHING EXPERIENCE

**Texas A&M University**, College Station, TX

- **Graduate/Undergraduate Mathematics and Probability Grader** Sep. 2023 – May 2024
  Evaluated mathematical solutions and provided technical feedback for undergraduate and graduate-level coursework.

**National Tsing Hua University**, Hsinchu, Taiwan

- **Teaching Assistant, Calculus** Sep. 2021 – Jun. 2022
  Led tutorial sessions and explained analytical and computational problem-solving techniques.
- **Teaching Assistant, Complex Analysis** Feb. 2021 – Jun. 2021

## AWARDS

- **ITW Travel Grant**, IEEE Information Theory Workshop, 2026
- **ISIT Travel Grant**, IEEE Information Theory Society, 2025
- **Graduate Merit Scholarship**, Department of Electrical and Computer Engineering, Texas A&M University, 2023–2024
- **Merit Award**, ATIDA Implementation Project Contest, 2021
- **Honorable Mention**, NTHU Electrical Engineering Undergraduate Project Competition, 2021